CST1500 – Laboratory Worksheet - Week 4

This week will serve multiple purposes and is slightly different from the previous 3 weeks. Week 5 will resume the format from the past 3 weeks.

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Week 4 Schedule

1. Week 4 will be used as a catch up for the lab exercises and lectures. Please try to make sure you have watched all of the lectures from Lecture 01 – Lecture 09 and have read through and completed the exercises from week one to week three inclusive.

2. If you have completed everything, you may continue to the next lab below - An Introduction to Batch Files and Scripting in Windows.

3. If you are up to date or when you have caught up, you can use this week to work on Coursework One.

4. You may take the FORMATIVE Test 1 to test yourself on what you have covered in the past 3 weeks. It covers material from the first 6 lectures (Lecture 01 – Lecture 06 inclusive). This would be useful for your own development as well as serving as an indicator of your progress to the teaching team. Your lab tutor will provide you with the specified time for the test in the lab.

An Introduction to Batch Files and Scripting in Windows

Lab Objectives

In this session, you will be introduced to some of the facilities that Windows provides for scripting. Whilst the scripting languages supported by Windows are, in many cases, quite different from those supported by Unix, the underlying purpose is the same – to automate repetitive administrative tasks such as file backups, account creation and software installation.

History of the Command-Line Language

Before there was Windows... there was the Microsoft Disk Operating System commonly referred to as MS-DOS. There was also a popular operating system called UNIX, but it was more popular with scientists and super computer users, while many of the “ordinary folk” used MS-DOS. Whereas Windows has a lovely graphical user interface, DOS was all done with text commands. That is, you typed out what you wanted the computer to do. Later the Windows operating system was built on top of DOS, so that DOS commands would run in the background, corresponding to how one interacted with the graphical interface. For example, if you click and dragged a file to a folder to move it, Windows would literally run the corresponding move command in DOS.

Windows Command Prompt

So, how do we start giving text commands to Windows? First, we need to open the command prompt (prompt for short or CMD). Click on the Windows Start Button, then type “cmd” in the Search box, and click on the program cmd.exe that appears in the search list. A cheery black and white window should pop up. Notice: there are no menus! Click in the Command prompt window to begin entering commands.

Batch Files

Prior to Windows 98, the only scripting language supported by default in Windows was the MSDOS batch file language. Although significantly more powerful scripting languages are available in current versions of Windows (see below for details), batch files can still provide useful functionality. As you would expect, the batch file syntax is generally much easier to learn than the more powerful scripting languages such as VBScript and Power-shell.

Hence, batch files are files that contain a list of commands. It executes the commands within the file when you execute the batch file. The commands execute one by one. A command within a batch file can be used to start/run another application, for example you could start MS Word.

Why batch files are useful?

A batch file could be used to run frequently run commands, deleting a series of files, moving files, copying files etc. A simple batch file does not require any special programming skills and can be done by users who only know DOS commands.

An example of a well-known batch file is the autoexec.bat, which is a simple batch file loaded each time the computer is loaded on MS-DOS and early Windows computers.

This batch file contained all the necessary commands and programs used to run MS-DOS and Windows each time the computer booted.

Ref for further examples and information:
www.chebucto.ns.ca/~ak621/DOS/BatBasic.html

EX 1: Getting started with Batch files

Creating Batch Files

Below are some examples of batch files. In each case, you should

1. Write the batch file using Windows Notepad.
2. Save the batch file(s) in your working directory. Note that batch files always have an extension of (.bat). When saving (and running) your batch files, make sure that your working directory is one in which you have the necessary permissions to create files and directories.
3. Run the batch file by typing its name (without the .bat extension) at the command prompt. Note that in the case of makedir.bat, you will need to provide the batch file with a parameter. In the first instance, use test i.e. makedir test. In the case of parent.bat and child.bat, you need to create both batch files and save them in your working directory before you can run parent.bat.
4. Work out what the batch file does and how it does it.

Note that you can find a lot of useful information about batch files, including their syntax, by pressing F1 on the Windows desktop and searching on the words batch files.

makedir.bat

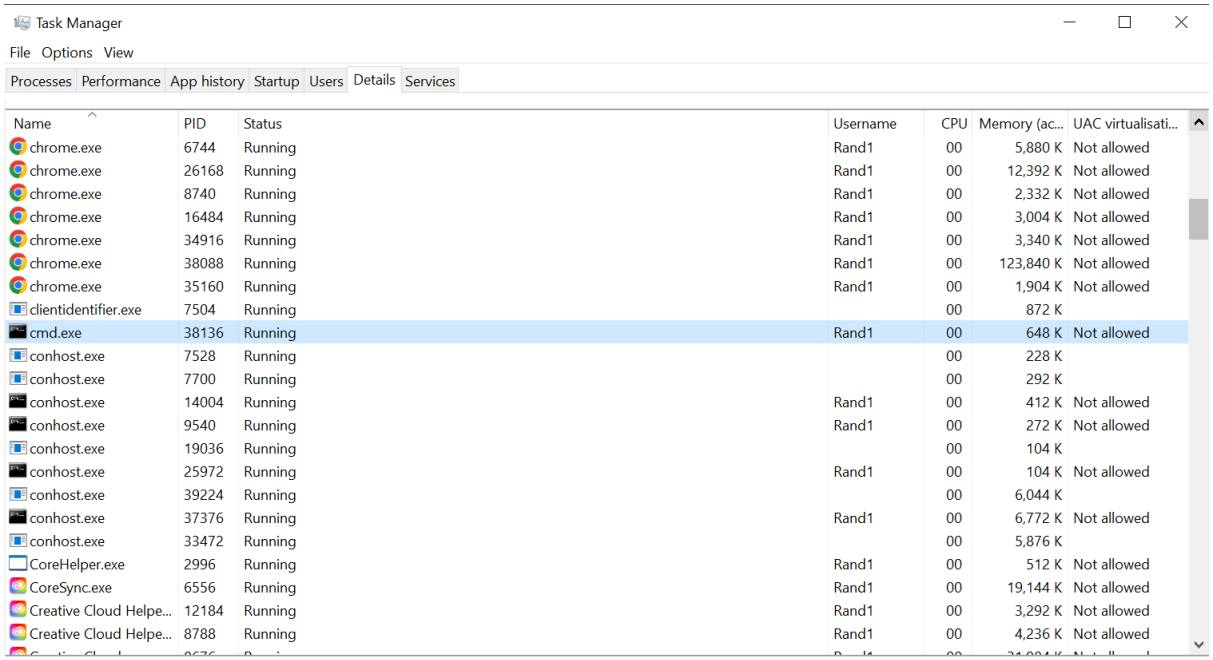
```
@echo off
rem This batch file demonstrates the use of selection and
rem the substitution of variables from the command line
rem
rem Run this script using the command makedir parameter
rem where parameter is the name of the directory to be created.
rem
if "%1"==”” goto skip
mkdir %1\dir1
mkdir %1\dir2
mkdir %1\dir3
echo some text > %1\dir1\file1.txt
echo more text > %1\dir2\file2.txt
echo yet more text > %1\dir3\file3.txt
goto end
:skip
echo No directory specified
:end
```

What is the significance of the rem command and the variable %1? What are the Unix equivalents of rem and %1? Make some notes.

EX 2: Understanding batch code of parent/child relationship

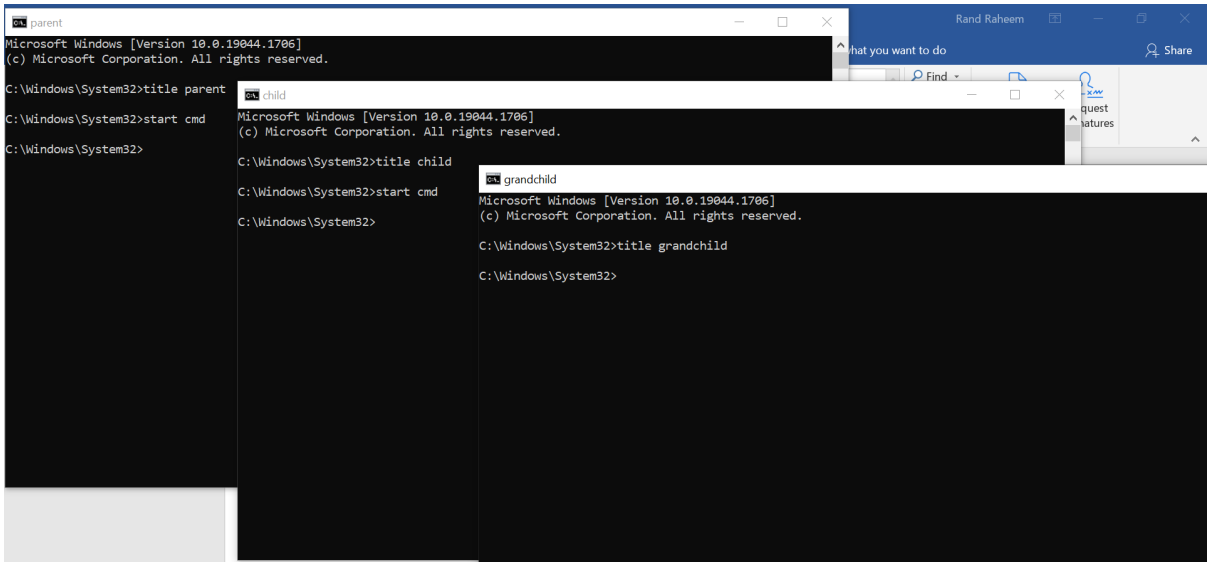
To simply understand the parent/child relationship, start by opening a cmd window, then give it a title e.g. parent - that is because will be opening more cmd windows and we do not want to lose track of them all. Type in **title parent** and press enter.

You need to check your process ID (PID) by opening the Windows Task Manager. Press **ctrl**, **alt** and **delete** together, then select **Task Manager**. Select the **Details** tab in the task manager. Make a note of the PID – it is important.

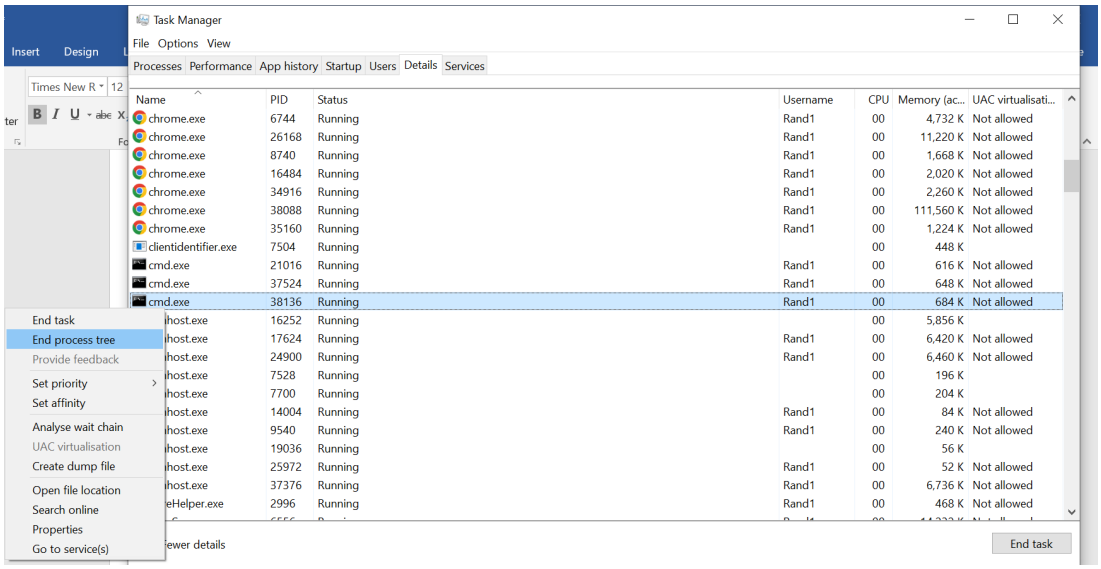


Name	PID	Status	Username	CPU	Memory (ac...	UAC virtualisati...
chrome.exe	6744	Running	Rand1	00	5,880 K	Not allowed
chrome.exe	26168	Running	Rand1	00	12,392 K	Not allowed
chrome.exe	8740	Running	Rand1	00	2,332 K	Not allowed
chrome.exe	16484	Running	Rand1	00	3,004 K	Not allowed
chrome.exe	34916	Running	Rand1	00	3,340 K	Not allowed
chrome.exe	38088	Running	Rand1	00	123,840 K	Not allowed
chrome.exe	35160	Running	Rand1	00	1,904 K	Not allowed
clientidentifier.exe	7504	Running		00	872 K	
cmd.exe	38136	Running	Rand1	00	648 K	Not allowed
conhost.exe	7528	Running		00	228 K	
conhost.exe	7700	Running		00	292 K	
conhost.exe	14004	Running	Rand1	00	412 K	Not allowed
conhost.exe	9540	Running	Rand1	00	272 K	Not allowed
conhost.exe	19036	Running		00	104 K	
conhost.exe	25972	Running	Rand1	00	104 K	Not allowed
conhost.exe	39224	Running		00	6,044 K	
conhost.exe	37376	Running	Rand1	00	6,772 K	Not allowed
conhost.exe	33472	Running		00	5,876 K	
CoreHelper.exe	2996	Running	Rand1	00	512 K	Not allowed
CoreSync.exe	6556	Running	Rand1	00	19,144 K	Not allowed
Creative Cloud Helpe...	12184	Running	Rand1	00	3,292 K	Not allowed
Creative Cloud Helpe...	8788	Running	Rand1	00	4,236 K	Not allowed

Now create a child process by typing **start cmd** in the parent window and press enter, name this window child, then create a grandchild process by typing **start cmd** in the child window and press enter and call it grandchild.



All three process PIDs can be checked via your Windows Task Manager – collect the PIDs as you create the cmds, as it is not easy to differentiate between them in the task manager. Now if you remember which cmd is your parent process tree, right click on it and then click on End Process Tree - that will end the entire tree. In other words, if you kill the parent process, the entire process generation will be killed as well.



Now if you repeat the same procedures in terms of creating the parent, child and grandchild processes, then you kill the child process what would happen? Explain why that is happening. Why are the parent and grandchild still there? Now if you kill the parent, why is the grandchild still there by itself? What do you know about Zombie processes?

Let us get into more details about how to create batch parent/child processes and to understand the relationship between them.

Parent.bat and child.bat

parent.bat
```
@echo off
for %%j in (1,2,3,4,5) do call child.bat
```

child.bat
```
set /p inputvar=Please type option #%j%:
echo You typed %inputvar% - Thank you.
```

In the above example, what is the precise relationship between parent.bat and child.bat? Also, what values does parent.bat pass to child.bat and vice versa? Explain your output to your lab tutor, SLA or classmate.

Another example of the parent.bat and child.bat

parent.bat
```
@echo off
rem This file demonstrates how one
rem batch file can call another cls
call child.bat first
set first=%inputvar%
call child.bat second
set second=%inputvar%
echo You typed %first% and then you typed rem %second%
```

child.bat
```
rem This batch file demonstrates how input
rem can be obtained from the user
set /p inputvar=Please type the %1 option:
echo Thank you
```

What is the precise relationship between parent.bat and child.bat? What values does parent.bat pass to child.bat and vice versa? Make some notes.

EX 3: Understanding batch code of looping mechanism

loop.bat
```
@echo off
rem This batch file demonstrates the use of loops. For the third
rem loop to work you need to enter 1 or more parameters e.g.
rem loop parameter1 parameter2 parameter3
rem
for %%i in (rock paper scissors) do echo %%i
rem
for /l %%n in (0,10,100) do echo %%n
rem
:loop
if "%1"==”” goto end
for %%c in (%1) do echo %%c shift
goto loop
:end
pause
```

What is the significance of `%%i`, `%%n` and `%%c`? What is the purpose of the shift and pause commands?

The Windows Script Host

One of the features that traditionally distinguished Unix and Linux from Windows was the availability of powerful scripting languages such Bash, Perl and Python. Originally released as part of Windows 98, the Windows Script Host was Microsoft's attempt to address the absence of powerful scripting facilities in Windows and represented a replacement for the rather limited batch file language.

As its name implies, the Windows Script Host (WSH) hosts a language and provides a shell within which scripts can run. Scripts can be created in the language of your choice provided that it is supported. By default, WSH supports VBScript and JScript but a variety of additional languages, such as Perl, Python, Tcl and REXX, are also available. If you want to learn a little bit about Perl (or some of the other freely available WSH-compatible scripting languages), go to www.activestate.com from where you can download the Perl software for Windows and access a significant amount of documentation and code examples.

Windows supplies two programs used to run WSH scripts. Wscript.exe is used to run WSH scripts from within the GUI environment whilst cscript.exe is used to run them from the command line or within batch files.

EX 4: Creating and understanding WSH Scripts

Below are some examples of VBScripts. In each case, you should:

1. Write the script using Windows Notepad.
2. Save the scripts in your working directory. Note that VBScripts always have an extension of .vbs.
3. Run the script by i) double-clicking on its icon in Explorer or ii) typing at the command prompt, cscript scriptname.vbs.
4. Work out what the script does and how it does it.

Note that you can find a lot of useful information about WSH by pressing F1 on the Windows desktop and searching on the word WSH. Additional information and numerous code samples are available at:
http://www.microsoft.com/technet/scriptcenter/scripts/default.mspx?mfr=true

helloworld.vbs

```
'this is my first VBScript
Wscript.echo "Hello World"
```

If when you run helloworld.vbs from the command prompt, you get an error to the effect that the script cannot be found, you are in the wrong directory! You can either:

i) change directory so that the script is in the current directory
ii) specify the full path of the script e.g.
cscript "c:\documents and settings\michael\my documents\bin\helloworld.vbs"

iii) add the directory that contains the script to your path

Enter **path /?** to find out how to add directories to the current path.

EX 5: Understanding the VBScripts I

What happens to the output from helloworld.vbs when you run it:

i) by double-clicking on its icon in Explorer?
ii) by using cscript on the command line?

Are there any advantages to running scripts from the command line rather than via the GUI?

EX 6: Understanding the VBScripts II

Discuss with your tutor the expected output and what details the following scripts provide you with?

Processor.vbs

```
strComputer = "."
Set objWMIService = GetObject("winmgmts:\\" & strComputer &
"\root\CIMV2")
Set colCSes = objWMIService.ExecQuery("SELECT * FROM
Win32_ComputerSystem")
For Each objCS In colCSes
     WScript.Echo "Computer Name: " & objCS.Name
     WScript.Echo "System Type: " & objCS.SystemType
     WScript.Echo "Number Of Processors: " &
     objCS.NumberOfProcessors
Next
Set colProcessors = objWMIService.ExecQuery("Select * from
Win32_Processor")
For Each objProcessor in colProcessors
     WScript.Echo "Manufacturer: " & objProcessor.Manufacturer
     WScript.Echo "Name: " & objProcessor.Name
     WScript.Echo "Description: " & objProcessor.Description
     WScript.Echo "Processor ID: " & objProcessor.ProcessorID
     WScript.Echo "Address Width: " &objProcessor.AddressWidth
     WScript.Echo "Data Width: " & objProcessor.DataWidth
     WScript.Echo "Family: " & objProcessor.Family
     WScript.Echo "Maximum Clock Speed: " &
     objProcessor.MaxClockSpeed
Next
```

Keyboard.vbs

```
strComputer = "."
Set objWMIService = GetObject("winmgmts:" _
& "{impersonationLevel=impersonate}!\\" & strComputer &
"\root\cimv2")
Set colItems = objWMIService.ExecQuery("Select * from
Win32_Keyboard")
For Each objItem in colItems
     Wscript.Echo "Caption: " & objItem.Caption
     Wscript.Echo "Description: " & objItem.Description
     Wscript.Echo "Device ID: " & objItem.DeviceID
     Wscript.Echo "Is Locked: " & objItem.IsLocked
     Wscript.Echo "Layout: " & objItem.Layout
     Wscript.Echo "Name: " & objItem.Name
     Wscript.Echo "Number of Function Keys: " &
     objItem.NumberOfFunctionKeys
     Wscript.Echo "Password: " & objItem.Password
     Wscript.Echo "PNP Device ID: " & objItem.PNPDeviceID
Next
```

EX 7: Understanding the VBScripts III

Earlier you were asked if there were any advantages to running scripts from the command line rather than via the GUI. On the basis of your experience with keyboard.vbs, would you change your mind? Whatever the answer, give reasons.

EX 8: Perform administrative tasks using Windows Command Line (Task 1)

Using the Windows Command Line create a folder on your desktop called “CST1500”, inside that folder create three more folders called “a”, “b” and “c”.

Store a hostname in a text file called “hostname.txt” in folder “a”.

EX 9: Perform administrative tasks using Windows Command Line (Task 2)

Store a structure of files in the C:\ drive in a file called “c_files_structure.txt” in folder “b” (tip you have to use “dir” command).

EX 10: Perform administrative tasks using Windows Command Line (Task 3)

Move folders “a” and “b” with their content to folder “c”.

EX 11: Perform more complex administrative tasks using Batch scripting

i. What is the purpose of the Path command?
ii. Write a Batch script that checks if a path given as a parameter exists. If it does, then it should print a message to the user informing them of the result. If it doesn’t work, then it should create the path and inform the user that the path has been created successfully.